Venkata Lakshmi Parimala Pasupuleti

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Virginia, USA (open to NYC relocation)

PROFESSIONAL SUMMARY

Full-stack software engineer and data engineer with 8+ years building production-grade systems for housing operations -- the exact domain EliseAI is automating. Startup mindset, bias for action, strong ownership: at Entrata I took on significant ownership of data pipelines, ETL infrastructure, and full-stack modules powering autonomous building operations for 100K+ monthly active users -- improving scalability and reliability at each layer. Delivered large-scale ETL (few thousand federal rows → 180,000 domain records), RabbitMQ event-driven async processing, machine learning pipelines, client-facing reporting, and 20+ end-to-end product features across 120+ Agile sprints. Strong in Python, Java, TypeScript, React, PostgreSQL, and AWS. Now building production agentic AI: LeaseI is an 8-agent LangGraph pipeline for housing compliance (93.6% precision on 1,240 test cases) and AgentProbe is a CI-integrated AI evaluation toolkit. M.S. Data Analytics Engineering, George Mason University (GPA 3.87). AgentProbe: github.com/venkatalakshmiparimala/AgentProbe

CORE COMPETENCIES

Autonomous Building Systems | Full-Stack (TypeScript / React / Python) | Large-Scale ETL & Data Pipelines | Data Modeling & Database Design | Data Quality Monitoring | Client-Facing Reporting Solutions | Microservices & Event-Driven Architecture | Agentic AI & Multi-Agent Orchestration

WORK EXPERIENCE

Blue Horse Digital
Sep 2026 -- Present

AI/ML Engineer (Volunteer) -- Virginia, USA

- Building ML pipelines and data workflows; integrating data systems with back-end and client-facing APIs; developing modular components across the full stack; surfacing metrics datasets and dashboards from raw business data.

Magizhs Tech Solution
May 2025 -- Aug 2025

AI/ML Engineer Intern (Forward Deployed)

- Owned product features end-to-end from design to deployment: built a LangGraph agentic pipeline with event-driven validation, direct API integrations with client enterprise systems, and client-facing outputs; served as primary technical consultant during rollout. Turned implementation patterns into reusable modular components handed off to the core engineering team.

George Mason University
Aug 2024 -- May 2026

Research Associate + Graduate Teaching Assistant

- Built multi-source prediction pipelines (Python, R, $R^2 = 0.977$); applied Isolation Forest for automated data quality checks across 3 research pipelines; designed Tableau and ggplot2 dashboards (+40% comprehension); fine-tuned a custom classification model (HuggingFace Transformers + PyTorch), deployed inference API on Quart.

Entrata (Housing Technology SaaS -- 100K+ monthly active users)
Jan 2017 -- Aug 2024

Senior Software Engineer -- Lehi, UT (Remote) [promoted from Associate, 2018]

- **Built and owned software enabling autonomous building operations for 7 years** -- 20+ production-grade full-stack modules (Python, PHP, JavaScript/TypeScript, React, PostgreSQL) across leasing workflows, eligibility calculations, rent and HAP processing, maintenance operations, and resident communication systems. Owned each feature end-to-end from architecture and design patterns through deployment, observability, and long-term maintenance. 99.5% uptime, zero compliance violations.
- **Designed and maintained large-scale ETL and data pipelines** with full data quality monitoring: annual HUD income-limit ETL (few thousand federal rows → ~180,000 domain records); RabbitMQ event-driven async processing with idempotency and retry guarantees; AWS Glue ETL jobs; data modeling and schema design for 20+ production projects across multi-tenant PostgreSQL environments. Built data quality checks and reconciliation processes -- one bad import affects thousands of residents' eligibility decisions.
- **Developed client-facing reporting solutions and metrics dashboards** powering both internal decision-making and customer-facing compliance workflows: 56 state-specific housing certification documents with merge fields; financial reconciliation reports; cross-functional dashboards surfacing eligibility, portfolio, and operational metrics. Implemented real-time data streaming via RabbitMQ and Kafka-compatible pipelines; maintained Snowflake-compatible data warehouse schemas.
- **Built scalable, reliable infrastructure:** Docker microservices redesign (50% scalability gain, 30% deployment time reduction); 200+ SQL query optimizations (30% performance gain) via execution plan analysis, indexing, and partitioning; CI/CD pipelines via GitHub Actions; observability tooling cutting mean debug time 40%; AWS S3 migration (100% PHI compliance, zero security incidents). Mentored 5+ engineers via 300+ code reviews; eliminated 80% of manual operational workload through Python/Bash automation.

PROJECTS

Leasel: Agentic AI Platform for Housing Compliance Operations

LangGraph 8-agent -- FastAPI -- RAG -- Python -- Housing AI

Production AI system for the same housing domain EliseAI operates in: 8-agent LangGraph pipeline (Triage → Retrieval → Configuration → Historical → Structured Data → Reasoning → Validation → Human Escalation) with durable state management, hybrid TF-IDF + semantic RAG, citation grounding, and HITL escalation. Core design rule enforced in code: LLM reasons; deterministic logic or a human makes every final compliance call. Results: 93.6% precision / 87.9% F1 on 1,240 test cases; 100% citation correctness. Built end-to-end with full product ownership and no PM.

github.com/venkatalakshmiparimala/Lease_complaince_rag

AgentProbe: Open-Source AI Evaluation & Observability Toolkit

FastAPI -- LangGraph -- ChromaDB -- CI/CD

15-endpoint FastAPI backend with 5-metric eval engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression gate with golden datasets, multi-provider LLM router (OpenAI/Anthropic/Groq/Ollama), and observability dashboard. Measures whether deployed AI systems achieve their operational objectives -- used to evaluate Leasel across 1,240+ test cases. Zero vendor dependency, zero cost.

github.com/venkatalakshmiparimala/AgentProbe

AI on FHIR: NLP Integration Layer for Clinical Records Systems

Flask -- React -- LLMs -- BERT -- REST API -- Docker

Full-stack platform (Flask backend + React frontend) translating natural-language queries (EN/ES) into FHIR API calls using LLMs, BERT, and embeddings. Built modular frontend components with focus on user experience; REST API layer with HIPAA-compliant RBAC; webhook-driven event routing across multi-system data stores. ~60% reduction in manual retrieval time.

github.com/venkatalakshmiparimala/AI_on_FHIR

EDUCATION

M.S. Data Analytics Engineering -- George Mason University — May 2026

GPA 3.87 -- Virginia, USA

B.Tech Computer Science & Engineering -- JNTU Anantapur — 2017

SKILLS

Languages & Frameworks: Python, Java, TypeScript, JavaScript, React.js, Node.js, SQL, PHP, FastAPI, Flask, Django, Spring Boot, C#

Data Engineering: ETL Design & Implementation, Large-Scale Data Pipelines, Data Modeling, Dimensional Modeling, Data Quality Monitoring, Snowflake, PostgreSQL, MySQL, Oracle, Apache Airflow, DBT, Apache Kafka, RabbitMQ, Real-Time Data Streaming, Data Warehousing, Metrics Datasets & Dashboards, Tableau, Power BI

AI & Agents: LangChain, LangGraph, Agentic AI, Multi-Agent Orchestration, RAG Pipelines, LLMs, HITL, Observability, Prompt Engineering, Function Calling, AI Guardrails, MCP, OpenAI API, Anthropic API, HuggingFace, PyTorch

Infrastructure & Architecture: Microservices, Event-Driven Distributed Systems, Scalable Architecture, System Design, AWS (S3, EC2, RDS, Glue, Lambda, Step Functions), Docker, Kubernetes, CI/CD (GitHub Actions), Continuous Integration, Observability, PostgreSQL, Redis, Linux, Production Debugging

Housing Domain: Autonomous Building Operations, Eligibility & Rent Calculation Workflows, HUD Compliance, LIHTC / Tax Credit, Leasing Systems, HAP Calculations, Affordable Housing Operations, Client-Facing Reporting